Transfer: waves and signals

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Choose equations or diagrams that fit the wave. Explain what the signal tells you.

SUCCESS CRITERIA - I can...

1. Use wave speed and round-trip geometry correctly.
2. Separate frequency, amplitude and direction effects.
3. State the limits of an inference from a signal.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “frequency” and “wavelength” to explain your answer. Sentence starters are optional.

frequency: number of cycles each second

wavelength: distance between corresponding points on repeated waves

round trip: travel from start to a reflector and back

01 Warm-up: bring back an earlier idea

A radar echo’s complete outward-and-return path is **5000 m**. **Find the target range** and **explain the division**.

Sentence starter: I think … . I can point to …, which shows …, so … .

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: $v = f\lambda$; $T = \mathbf{1}/f$; $d_{range} = v\,\Delta t/\mathbf{2}$.
d_{range} is one-way range from round-trip time. Match speed and equation to the wave and medium.

Worked reasoning: An echo time measures range using travel speed, while a frequency shift can indicate radial motion. Different observations answer different questions about the same target.

Your turn: Draw a measurement plan labeling outgoing pulse, return pulse and the time interval.

Sentence starter: My design includes … . It does what is needed by … .

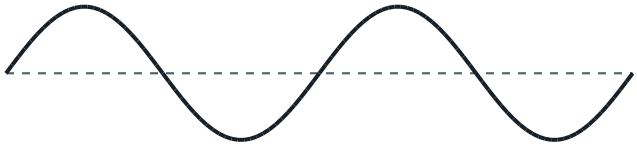


Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

1. **Sketch a wave** and **label amplitude** and wavelength **0**.
2. **Explain which tells about oscillation** size and which about spatial repetition.

Sentence starter: I think … . I can point to …, which shows …, so … .



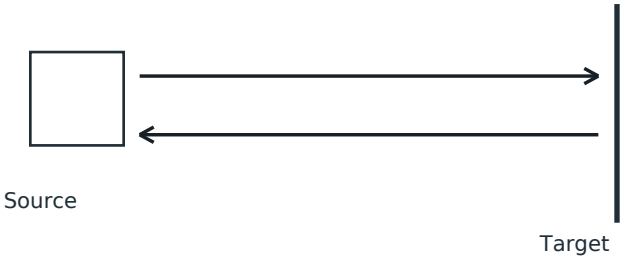
Sketch and label your response.

Marzano check - circle: ① ② ③

04 Apply the physics

A wave simulator uses frequency **5 Hz** and wavelength **2 m**. **Find propagation speed**. An echo using that same speed returns after **4 s**. **Label the round trip** and **calculate the target range**.

Sentence starter: I found … [unit]. This value means … in this situation.



Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Explain your reasoning

A source gets louder while its frequency remains fixed. Does pitch necessarily rise? Explain using wave properties.

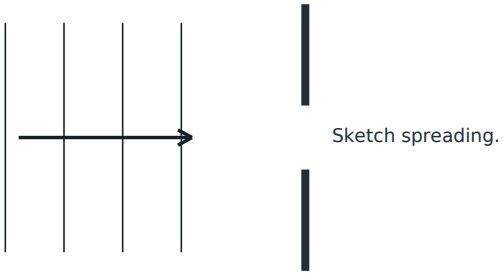
Sentence starter: I think … . I can point to …, which shows …, so … .

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

Two waves meet the same opening with wavelengths **4 cm** and **6 cm**. Which generally diffracts more? **Explain the relevant comparison**.
Label wavelength and opening width; **sketch the emerging wavefronts**.

Sentence starter: I think … . I can point to …, which shows …, so … .

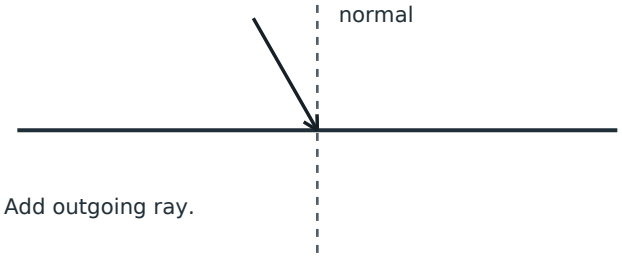


Marzano check - circle: 1 2 3

07 Exit check

A ray enters a slower medium at an angle. Sketch whether it bends toward or away from the normal and **name what happens** to frequency.

Sentence starter: In my work, I used … to show … . I labeled it … .

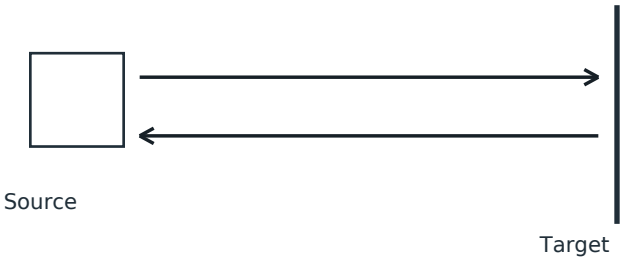


Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

A sound echo arrives after **0.5 s** and a light echo after **2 microseconds**.
1. Use **340 m/s** and **3×10^8 m/s**.
2. Which target is farther?

Sentence starter: I notice … . This supports … because … .



Marzano check - circle: 1 2 3